

Ranking Papers by their Short-Term Scientific Impact

Ilias Kanellos¹, Thanasis Vergoulis¹, **Dimitris Sacharidis**², Theodore Dalamagas¹, Yannis Vassiliou³



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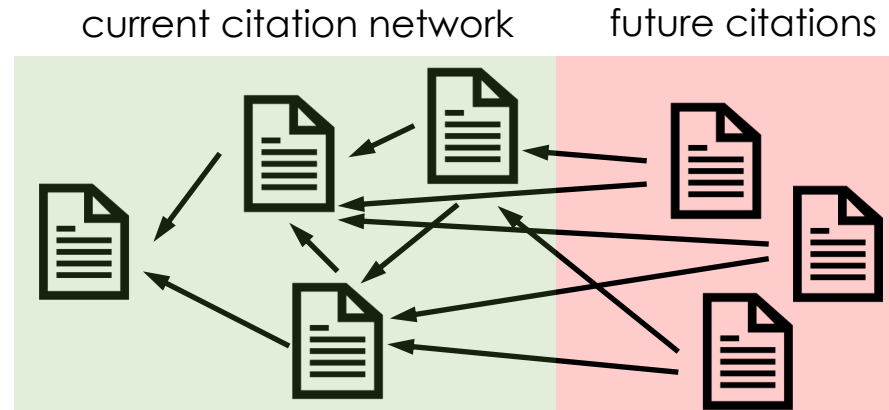
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National Technical
University of Athens

Scientific Impact of Papers

citation count = number of citations a paper receives
= in-degree in the citation network



short-term impact = number of future citations a paper receives

problem: rank **current papers** based on their **future citations**

Citation Network Evolution

Assumption 1: PageRank at time t indicates the chance a paper will get cited at time $t + 1$.

$$\mathbf{v}^{(t)} = \alpha \mathbf{S}^{(t)} \mathbf{v}^{(t)} + (1 - \alpha) \mathbf{u}^{(t)}$$

PageRank

stochastic citation matrix

teleport probability

random jump probability

Assumption 2: Future papers only cite current papers.

So, to compute a future PageRank at time $n + t$, **contract** network back to current time n and **modify teleport probabilities**

$$\hat{\mathbf{u}}_{[n]}^{(n+t)} = \mathbf{u}_{[n]}^{(n+t)} + \alpha \sum_{i=1}^t u_{n+i}^{(n+t)} S_{[n],n+i}^{(n+t)} = \mathbf{u}_{[n]}^{(n+t)} + \mathbf{w}_{[n]}^{(n+t)}$$

original teleport prob.

additional teleport prob.

Computing Short-Term Impact

So, from Assumption 1:

$$\text{short-term impact} \propto \mathbf{y} = \sum_{t=0}^{\tau-1} \mathbf{v}_{[n]}^{(n+t)} \quad \text{future PageRanks}$$

which can be computed as a PageRank from Assumption 2:

$$\mathbf{y} = \alpha \mathbf{S}^{(n)} \mathbf{y} + \beta \mathbf{w} + \gamma \mathbf{u}^{(n)}$$

for non-negative $\alpha + \beta + \gamma = 1$

short-term impact

stochastic citation matrix

teleport probability

attention probability

where the **attention** probability $\mathbf{w} \equiv \sum_{t=0}^{\tau-1} \mathbf{w}_{[n]}^{(n+t)}$ is estimated from history

Evaluation

Datasets

- **APS**: collection of ~500 000 physics papers from American Physical Society
- **DBLP**: collection of ~3 million computer science papers

Protocol

- Partition datasets in two halves
 - **Current state** of citation network is the older half
 - **Future state** of varying length (in terms of num. of papers) from newer half
- Vary **test ratio** = $\frac{\#papers\ in\ future}{\#papers\ in\ current}$
- Ground truth ranking from future state

Measures

- Spearman's ρ correlation
- nDCG@50

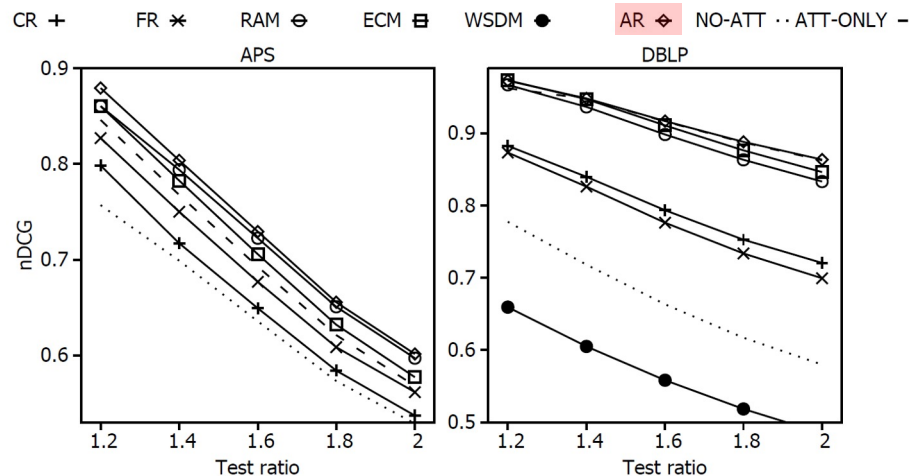
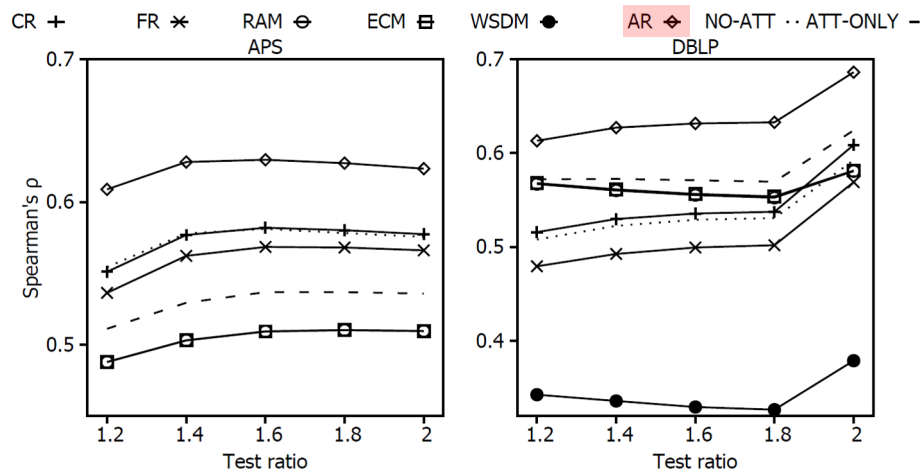
Methods

- our approach, Attention Rank (AR)
- our approach **without** attention (NO-ATT)
- our approach **only-with** attention (ATT-ONLY)
- CiteRank (CR)
- FutureRank (FR)
- Retained Adjacency Matrix (RAM)
- Effective Contagion Matrix (ECM)
- Winner of WSDM'16 Cup (WSDM)

Results

Attention Rank outperforms competitors, particularly in Spearman's ρ

Attention is an effective mechanism, but not on its own



Thank you!

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